

Chapter 115 — Electrical Stimulation and Neuromodulation in Storage and Emptying Failure — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Sacral nerve stimulation (SNS)** = **third line** for idiopathic overactive bladder (OAB): American (AUA) and European (EAU) guidelines agree.
- **SNS test positive:** **≥50% improvement** in ≥1 bladder-diary parameter.
- **Retention test positive:** catheterisations ↓ **≥50%**; post-void residual (PVR) = only objective marker.
- **S3 target:** anal bellows + great-toe flexion; **S2** calf/foot rotation; **S4** no toe.
- **FDA:** urge urinary incontinence (UUI) 1997; urgency-frequency + non-obstructive retention (NOUR) 1999.

- **Not FDA approved:** bladder pain, pelvic pain, neurogenic bladder, chronic pudendal stimulation.
- **SUMiT:** percutaneous tibial nerve stimulation (PTNS) 55% vs sham 21%.
- **PTNS:** 34 G needle, 3–4 cm above medial malleolus; 12 × 30 min.
- **ROSETTA:** onabotulinumtoxinA (BTX) 200 U ≈ SNS at 2 years; more urinary tract infection (UTI), clean intermittent catheterisation (CIC).
- **Detrusor overactivity (DO) does not predict SNS success.**

HISTORY OF ELECTRICAL STIMULATION

- **CE 46 Scribonius Largus:** torpedo fish for gout, headache.
- **Direct organ stimulation failed:** painful, only local → afferent (indirect) route.
- **Brindley sacral anterior root stimulator (SARS):** anterior roots → voiding, defecation, erection; + **dorsal rhizotomy** for DO.
- **≈1985 sacral neuromodulation:** Tanagho and Schmidt; decreases bladder activity.
- **McGuire 1983:** first tibial nerve stimulation for bladder.

PATHOPHYSIOLOGY OF STORAGE AND VOIDING DISORDERS

- **Exclude anatomy first:** stones, cancer, fistula, stricture, prolapse, enlarged prostate.
- **Pontine storage centre = L-region**, lateral; **pontine micturition centre (PMC) = M-region**, medial.
- **Voiding:** external sphincter relaxes → **1–2 s** → detrusor contracts.
- **Sphincter relaxation:** PMC → γ-aminobutyric acid (GABA)/glycine interneurons.
- **Sympathetic fires only from ≈60% capacity;** not main continence mechanism.
- **Forebrain lesion:** timing lost, reflex intact.
- **Filling afferents Aδ;** OAB reflex via **C fibres** (naloxone-sensitive).

WORKING MECHANISM OF ELECTRICAL STIMULATION OF THE LOWER URINARY TRACT

- **Current view:** afferents → forebrain awareness/alertness → ↓ PMC firing.
- **SNS recruits ≥1000× more axons**, continuously.
- **Underactive bladder:** SNS works mainly via **sphincter relaxation**.

INTRAVAGINAL ELECTRIC STIMULATION

- **For stress incontinence;** safe; evidence low quality.

TRANSCUTANEOUS SACRAL DERMATOME STIMULATION

- **Paediatric S3 dermatome:** 10 Hz, 20 min daily → less incontinence, UTI, PVR.
- **S3 dermatome:** ↑ capacity; **cannot acutely suppress DO** (conditional fails).

PUDENDAL NERVE STIMULATION

- **S2–S4**, Alcock's canal; off-label tined lead, **posterior approach** (perineal → infection).
- **Salvage after failed SNS;** not FDA approved.

- **Indications:** idiopathic DO, neurogenic DO, chronic pelvic pain.

DORSAL GENITAL NERVE STIMULATION

- **Experimental;** purely sensory; confirm by **genitoanal reflex**.

TRANSCUTANEOUS TIBIAL NERVE STIMULATION

- **PTNS FDA 2000, National Institute for Health and Clinical Excellence (NICE) 2006.**
- **Transcutaneous (TTNS) < percutaneous:** skin capacitor. TTNS ≈ PTNS ≈ pelvic floor training; < **antimuscarinics**.

PERCUTANEOUS TIBIAL NERVE STIMULATION

- **Tibial fibres S2–S3;** lower urinary tract innervation L2–S4.
- **SP-6:** 6–8 cm above medial malleolus; electroacupuncture 2–15 Hz, 10–20 mA.
- **Settings:** 9 V, 0–10 mA, **200 μs**, **20 Hz**; needle 60–90°.
- **Motor:** toe flexion/fanning; **sensory:** sole tickling.
- **Objective success:** urgency/UUI ↓ **≥50%** + frequency ↓ **≥25%**.
- **Outcomes:** subjective ≈60%; diary 47–56%; **placebo ≈21%**.
- **SUMiT:** n = 220, first validated sham, 12 weeks.
- **SUMiT extension:** 14-week taper → **77% relapse-free at 3 years**.
- **Durability:** 53–77% at 2–3 years **only with maintenance**.
- **vs tolterodine:** equal efficacy, fewer side effects.
- **Cochrane:** antimuscarinics first; stimulation for **antimuscarinic-refractory**.
- **Children:** PTNS after bladder training, pelvic floor therapy, antimuscarinics fail.
- **Regimen:** 12 weeks best supported; 1–3 sessions/week.
- **Prognosis:** better without DO; low Short Form-36 (SF-36) mental score → failure.
- **Implant trap:** PTNS response ≠ implant response.
- **Implants:** Urgent-SQ first (2006), ≈5 cm above malleolus; eCoin primary battery; Protect-PNS wireless.

SACRAL NERVE STIMULATION

- **InterStim Conformité Européenne (CE) mark 1994;** rechargeables FDA 2019.
- **OAB definitions:** normal ≤7 voids/24 h (1.5–2 L); **OAB-dry ≥8 + urgency**.
- **Frequency-volume diary first** — excludes polydipsia.
- **InSite:** SNS 61% vs medical 42% at 6 months.
- **ROSETTA 6 months:** UUI –3.9 (BTX) vs –3.3 (SNS).
- **ROSETTA 2 years:** UTI 24% vs 10%; CIC 6%; SNS revision 3%, removal 9%.
- **Registered BTX dose idiopathic OAB = 100 U** (> placebo at 4 months).

- **NOUR:** PVR often ≥ 100 mL; capacity > 500 mL.
- **Bladder pain:** offer SNS only with coexisting OAB (objective diary).
- **Neurogenic:** success comparable, evidence low, unapproved.
- **NOUR odds $\uparrow$:** women + psychiatric history; men + prior transurethral prostate resection/bladder neck incision; **age $\downarrow$.**
- **Test for all:** direct implant theoretically $\geq 90\%$ success, but test in practice.
- **Diaries:** 3–5-day baseline vs test-period diary.
- **Responses:** sensory before motor.
- **Percutaneous nerve evaluation (PNE):** 5–7 days; positive $\approx 60\%$ OAB; **not for NOUR.**
- **S3 landmarks:** 9 cm from coccyx tip, 11 cm from anal verge, 1–2 cm lateral.
- **Staged:** percutaneous since 2003; test ≤ 4 weeks; stage 2 at 2–4 weeks.
- **Staged > PNE** conversion to implant; PNE lead dislocates.
- **International Continence Society (ICS) consensus:** S3 preferred; responses on all 4 contacts, low amplitude.
- **Prophylaxis:** consider meticillin-resistant *S. aureus* (MRSA) cover; cefazolin + metronidazole.
- **Programming:** 210 μ s, 10–16 Hz, sensory threshold.
- **Reprogramming:** ≈ 3 in year 1, then ≈ 0.8 /year.
- **Battery:** non-rechargeable ≥ 10 years; rechargeable ≥ 15 years.
- **Cycling** (16 s on/8 s off; 8 h on/16 h off) = continuous.

- **Bilateral not superior** to unilateral $\rightarrow$ not standard.
- **Magnetic resonance imaging (MRI):** whole-body 1.5 and 3 T conditional; remove all old components at replacement.

CLASSIC TRAPS

- **Trap:** SNS works by pelvic floor contraction — **Actually:** afferent forebrain modulation.
- **Trap:** urodynamic DO needed for SNS — **Actually:** DO does not predict success.
- **Trap:** ROSETTA shows BTX beats SNS — **Actually:** used 200 U, twice registered dose; equal at 2 years.
- **Trap:** SNS approved for bladder pain or neurogenic bladder — **Actually:** OAB and NOUR only.
- **Trap:** PNE for retention — **Actually:** staged implant; PNE fails more.
- **Trap:** PTNS effect is permanent — **Actually:** relapse likely without maintenance.
- **Trap:** TTNS equals PTNS physiologically — **Actually:** skin impedance limits amplitude.
- **Trap:** sympathetic tone maintains continence — **Actually:** fires only from $\approx 60\%$ capacity.
- **Trap:** pelvic floor physiotherapy for retention — **Actually:** usually ineffective; tone returns to baseline.
- **Trap:** bilateral SNS after unilateral failure — **Actually:** not superior; not standard.